Kimberley Tanatswa Muchenje

Ph.D. Candidate | Division of Biology and Biological Engineering | California Institute of Technology
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SUMMARY

My research focuses on developing genome engineering technologies for precise**, targeted DNA insertion** in plants. I combine **transposon engineering** with **synthetic biology, protein and RNA engineering**, and **nanoparticle-based delivery** to expand the range of plants and genomic applications that can be engineered. These technologies have potential applications in climate-resilient agriculture, sustainable bioproduction, and fundamental plant biology. I am committed to **teaching, mentoring, and increasing access** to bioengineering education.

EDUCATION

California Institute of Technology, Pasadena, CA **March 2023 – Present**
Ph.D. Candidate, Division of Biology and Biological Engineering
Advisor: Gözde S. Demirer, Division of Chemistry and Chemical Engineering
Thesis: Engineering site specific transposons for targeted DNA insertion into plant genomes

Davidson College, Davidson, NC **August 2018 – May 2022**
B.S., Biology
Advisor: Dr. Debbie Thurtle-Schmidt, Biology Department

HONORS AND AWARDS

- Margaret McNamara Education Grant Fellow 2025
- Caltech 2-year fellowship for demonstrating commitment to Diversity and Inclusion in academia 2022
- Katherine Johnson award for STEM, Davidson College 2022
- Kathryn Davis Peace Project Grant Award 2022
- Alvarez Grant for Career exploration 2020
- Spencer Weinstein Travel Grant 2020
- Patterson Scholarship for International Women, Davidson College 2018
- Alvarez Scholarship for International students, Davidson College 2018
- USAP Zimbabwe Scholar for high achieving students across all high schools in Zimbabwe 2017

RESEARCH PROPOSALS & INDUSTRY COLLABORATIONS

BASF targeted genome editing open call, co-development proposal 2026
My proposal was selected for the second round after which, I presented it to their team and now await the next round of deliberation.

PUBLICATIONS

Peer-reviewed

- **Muchenje, K. T.**, McCombe, C. L., Wang, Y., Oz, T. M., Li, E., Saffron, A. & Demirer, G. S. Optimized R2 retroelement complexes for DNA insertion into plant genomes. *Nature Biotechnology* (2026). doi:10.1038/s41587-026-03197-y. First author.
- Li, E., Berruto, C., Oz, T. M., Grillo, E., Griffin, C., Wang, Y., **Muchenje, K. T.**, Jones, J. W. & Demirer, G. S. Sentinel plants enable quantitative monitoring of bioavailable nitrate in soils and microbial environments. *Proceedings of the National Academy of Sciences* 123, e2609666123 (2026). doi:10.1073/pnas.2609666123
- Li, E., Geng, Y., Khristoforova, T. R., Wang, Y., Jones, J. W., **Muchenje, K. T.**, Grillo, E., Zhang, H., Schmitz, R. J. & Demirer, G. S. Vacuum and sonication treatment enable efficient transient gene expression in various monocot and eudicot plant seedlings. *ACS Synthetic Biology* 15, 1452–1466 (2026). doi:10.1021/acssynbio.5c00855

Preprints

- Wang, Y., **Muchenje, K. T.**, Papikian, A., Legendre, M. G., Leem, E. & Demirer, G. S. CRISPR-associated transposases enable programmable DNA integration in plants. *bioRxiv* (2026). doi:10.64898/2026.07.15.738807
- Legendre, M. G., **Muchenje, K.T.**, Heredia, C. A., Colee, C. M., Wang, Y., Pistilli, V. & Demirer, G. S. Extracellular contractile injection systems for receptor-mediated protein delivery into plant cells. *bioRxiv* (2025). doi:10.1101/2025.08.22.671880. Fourth author.

PATENTS

- Demirer, G. S. & **Muchenje, K.** Optimized R2 retroelement complexes enable precise and efficient DNA insertion into plant genomes. Provisional patent CIT-9388-P, filed October 2025. Non-provisional U.S. application 30KJ-810033-US (2026).
- Demirer, G. S., Wang, Y. & **Muchenje, K.** Targeted DNA integration in plants by CRISPR-associated transposases (CASTs). Provisional patent CIT-9134-P, filed February 2024. PCT application CIT 9134-PCT / 30KJ-810004-WO.

RESEARCH EXPERIENCE

Graduate research: Demirer lab **March 2023 – Present**

Division of Chemistry and Chemical Engineering, California Institute of Technology

R2 retrotransposon-mediated DNA writing in plants

- Engineered the avian *Taeniopygia guttata* R2 non-LTR retroelement (R2Tg) into a double-strand-break-free editor for targeted insertion of 2.2–5 kb DNA payloads at the 25S rDNA safe-harbor locus in *N. benthamiana*, *A. thaliana*, and *S. lycopersicum*, achieving insertion in up to 24% of transformed cells and approximately one copy per genome by ddPCR, with stable copy number across payload sizes.
- Demonstrated 30-fold higher integration than Cas9-mediated homology-directed repair at the same locus.
- Characterized integration specificity and epigenetic outcomes using long-read chromosome mapping, tagmentation-based integration-site sequencing, and targeted bisulfite sequencing, observing 96% on-target integration, precise 5′/3′ junctions, and unmethylated payloads in moderately methylated rDNA.

CRISPR-associated transposase (CAST)-mediated DNA insertion in plants

- Adapted CRISPR-associated transposases from *Vibrio cholerae* (VchCAST) and *Pseudoalteromonas* (PseCAST) for targeted insertion in *N. benthamiana* leaves.
- Reconstituted PACE-evolved Type I-F evoCAST in plants, reaching 8% insertion of a 1 kb promoter payload at a bidirectional reporter landing pad.

Retargeting R2 editors to new genomic loci

- Extended R2 DNA writing beyond 25S rDNA by fusing the retroelement to SpCas9(H840A) nickase and reached 5% insertion by 5′-junction ddPCR at the 16C p35S-GFP locus in *N. benthamiana*.
- Screened R2 fusions with enOgeuIscB, ISDra2 eTnpBc, Cas12a, and SaCas9 nickases using large guide RNA libraries targeting multiple loci in *A. thaliana* and *N. benthamiana*.

Developing heritable R2-mediated insertion in tomato and Arabidopsis thaliana

- Discovered that constitutively active wild-type R2Tg is toxic to the Arabidopsis germline, with transformed seeds recovered only from reverse-transcriptase-dead and endonuclease-dead mutant lines.
- Built R2-editor expression systems using inducible 17β-estradiol, dexamethasone, and methoxyfenozide promoters and developed the lab's first successful Micro-Tom callus transformation protocol.
- Validated that inducible control of R2Tg enables transformation in dividing cells and developed pipelines toward heritable R2-mediated editing with editor expression and stability across generations in Arabidopsis and tomato.

R2 delivery using nanoparticles

- Delivered R2 editors via Photorhabdus virulence cassette (PVC) extracellular contractile injection system nanoparticles, observing mScarlet payload accumulation validating insertion events and measured 1.5% R2Tg integration after eCIS-mediated delivery.

Rotation: Stathopoulos lab **January 2023 – March 2023**

Biology and Biological Engineering, California Institute of Technology

- Used blue-light-controlled Dorsal LEXY nuclear export, FISH, and live imaging to investigate Dorsal-target gene expression from nuclear cycle 10 through gastrulation in *D. melanogaster*.

Rotation: Pachter lab **October 2022 – December 2022**

Biology and Biological Engineering, California Institute of Technology

- Developed a high-performing sequence-alignment tool for massively parallel reporter assays (MPRAs) using kallisto bustools and compiled an MPRA methods database for a planned review.

Independent study: Debbie Thurtle-Schmidt lab **2020 – May 2022**

Davidson College: Biology Department, Davidson, NC

- Characterized regulatory DNA binding by *C. elegans* mutant NHR-25(ku217) using ChIP-seq, CUT&RUN, and comparative sequence analysis of wild-type and mutant worms.
- Generated wild-type and mutant NHR-25–GFP fusions in *C. elegans* using CRISPR-Cas9 and worm microinjection.
- Redesigned an mKate-2/GFP reporter to study how the *blimp-1*/*cup-2* intergenic boundary element restricts NHR-25 regulation of *blimp-1*.
- Trained incoming independent-study students in research protocols and methods.

Conservation Biology Remote Research Fellow **May 2020 – June 2020**

Davidson College: Biology Department, Davidson, NC

- Analyzed biodiversity in 40 experimental grow bags following habitat loss and introduced species using R and Python, and presented species-accumulation, PCA, and NMDS results showing reduced richness.

RESEARCH RESOURCES

- R2Tg editor constructs deposited at Addgene, deposit #87796.
- NCBI SRA BioProjects PRJNA1463879 and PRJNA1484212.

MEDIA COVERAGE

"Improving Plants with a Tool Borrowed from Birds." *Caltech News*, June 2026. Interviewed as lead author.

TECHNICAL SKILLS

- **Synthetic biology and genome engineering**
- **Transformation and delivery:** Agrobacterium leaf and seedling transformation, protoplast isolation and PEG transfection, callus transformation and regeneration, nanoparticle delivery, *C. elegans* microinjection
- **Protein biochemistry:** recombinant protein expression, extraction and purification from plant fractions, bacteria and yeast, Western blotting, HiBiT split-luciferase assays
- **Genomic analysis:** ddPCR, junction PCR, RT-qPCR, single-cell genomics
- **Cellular analysis:** flow cytometry, confocal microscopy, image analysis (Fiji), FISH
- **Computational:** Python, R, bioinformatic data analysis, AlphaFold3, SnapGene, Geneious Prime, GraphPad Prism

PRESENTATIONS

- **Muchenje, K. T.** "Reprogrammable R2 retrotransposon platform for heritable targeted DNA insertion into plant genomes." **Talk** presented at Transposable Elements, Cold Spring Harbor Laboratory, NY, September 30, 2026.
- **Muchenje, K. T.** "Engineering jumping genes for targeted DNA insertion into plant genomes." **Invited speaker**, Davidson College Biology Department, Davidson, NC, October 2026.
- **Muchenje, K. T.** "Engineering site specific transposons for targeted DNA insertion into plant genomes." **Biology Seminar**, Division of Biology and Biological Engineering, Caltech, Pasadena, CA, January 2026.

- **Muchenje, K. T.**, McCombe, C. L., Wang, Y., Oz, T. M., Li, E., Saffron, A., Demirer, G. S. "High-efficiency Targeted Gene Insertion in Plants Using R2 non-LTR Retrotransposons." **Talk** presented at Plant Genomes, Systems Biology & Engineering, Cold Spring Harbor Laboratory, NY, September 2025.
- **Muchenje, K. T.**, McCombe, C. L., Wang, Y., Oz, T. M., Li, E., Saffron, A., Demirer, G. S. "Targeted DNA Insertion in Plants by Site-Specific Transposases." **Poster** presented at Synthetic Biology: Engineering, Evolution & Design (SEED), Houston, TX, June 2025.
- **Muchenje, K. T.** "Engineering site specific transposons for targeted DNA insertion into plant genomes." **Biology Seminar**, Division of Biology and Biological Engineering, Caltech, Pasadena, CA, October 2024.
- **Muchenje, K. T.**, Wang, Y., Demirer, G. "Transposon-Based Toolkit for Targeted DNA Insertion into Plant Genomes." **Poster** presented at the 7th International Conference on CRISPR Technologies, San Diego, USA, October 2024.
- **Muchenje, K. T.**, Thurtle-Schmidt, D. "Elucidating the effects of the *nhr-41(ok584)* mutation on the gene expression pattern in *C. elegans* to uncover the role of *nhr-41* during the L2 and L3d worm stages." **Poster** presented at the Verna Miller Case Symposium for Research and Creative Works, Davidson, NC, May 2022.
- **Muchenje, K. T.**, Thurtle-Schmidt, D. "Characterization of binding in mutant NHR-25, a transcription factor in *C. elegans* worms." **Poster** presented at the Summer Research Symposium at Davidson College, Davidson, NC, October 2021.

TEACHING AND MENTORING EXPERIENCE

Teaching Assistant, Bi/BMB 251b **2025 – Present**

Current Research in Biology and Biological Engineering, California Institute of Technology

- Support weekly Biology and Biological Engineering research seminars by preparing presenters, managing technical setup, and providing student presenters feedback on slides, delivery, and audience questions.

Teaching Assistant, ChE/Ch/BE 130 **Spring 2025 and Spring 2026**

Biomolecular Engineering Laboratory, California Institute of Technology

- Designed and taught a laboratory course using CRISPR-Cas9 to knock out genes in *N. benthamiana* and observe knockout phenotypes across two course offerings.
- Developed slides and protocols on experimental design, plant transformation, and genome editing, and led laboratory sessions and guided student experiments.

Instructor and Mentor, First-Year Success Research Institute (FSRI) **Summer 2024**

California Institute of Technology

- Mentored three incoming first-year students in an *E. coli* R2 retrotransposon expression and purification project, culminating in student poster presentations.

Jekesa Ramangwana, Founder **May 2022 – Present**

Davidson College Center for Civic Engagement, Harare, Zimbabwe

- Developed a genomics, bioinformatics, and genome editing course for high school and college students in Zimbabwe, covering applications in biomedical research, agriculture, and microbiology, supported by a $10,000 Kathryn Davis Peace Project award.
- Continue mentoring program alumni pursuing university study and research, while securing funding for annual in-person and virtual instruction.

Teaching Assistant **January 2023 – April 2023**

Foundational Principles in Molecular Biology, California Institute of Technology

- Led four weekly recitations and office-hour sessions for approximately 50 students in foundational molecular biology.
- Designed examination questions and graded assignments.

Mentees

Graduate students and rotation students

- Olivia Ettlin (2025), Erica Hengartner (2026), and Matthew Part (2026), rotation graduate students, on retargeting R2 editors to new loci.
- Matthew Ocanas (2025, R2 editors from low-temperature organisms) and Arsh Khetan (2025, viral-vector delivery of R2 editors), rotation graduate students.
- Sandra Temgoua (2025, rotation graduate student) and Aline Milach Teixera (first-year Ph.D. student), CAST-mediated insertion in plants.

Undergraduate and summer research students

- Lucas Garcia (Caltech SURF, 2026), heritable R2-mediated insertion in tomato and *Arabidopsis thaliana*.
- Indeever Madireddy and Paulina Chavira, first-year undergraduates, CAST-mediated targeted DNA insertion in plants.
- Amelia Saffron (first-year undergraduate), Gracy Wah, JingJing Duan and Talaysha Simonds (FSRI, 2024), R2 retrotransposon-mediated DNA writing in plants.

Prior to Caltech

Laboratory Assistant for 4 organic chemistry lab sections, Davidson College **2021 – 2022**

Teaching Assistant for the Liberal Arts Collaboration for Digital Innovation Data Science Course **2021**

Embedded Tutor for a 30-student introductory biology class under Dr. Kevin Smith **2020**

LEADERSHIP & INVOLVEMENT

Outreach and education

SynBio4ALL Africa, Instructor **December 2024 – Present**

Global, Virtual

- Connected synthetic biology expertise with African research interests and initiated a native-plant genome sequencing project to support region-specific research.
- Designed and delivered virtual synthetic biology curricula focused on plant genome editing to more than 100 students across Africa.

Research mentor, San Marino High School **September 2026 – Present**

San Marino, CA

- Mentor students biweekly throughout the school year through the Caltech STEM program on an 11 kb R2Tg payload carrying a four-gene mushroom luciferin biosynthesis pathway, testing full-length insertion and function.

Prior to Caltech

Impact and Engagement Assistant at the Ada Jenkins community center, Davidson, NC **2021 – 2022**

Student Leader, FIRST Action Team, Howard Hughes Medical Institute Inclusive Excellence Grant **2019 – 2022**

Residence Advisor for 25 students, Davidson College **2021 – 2022**

Student Volunteer for COVID-19 testing, Belvedere Hospital, Harare **2020 – 2021**

Student Intern building the GenZer mentoring program, Women of Color Advancing Peace and Security **2020**